Imran E Kibria

3^{rd} YEAR PHD RESEARCHER IN SPEECH INTELLIGENCE

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Research Interests

Speech Quality Assessment, Speech Quality Enhancement, Auditory Attention Detection, Audio Explainability.

Experience

Graduate Research Associate — *Columbus, OH*
Audio, Speech, & Perceptually-Inspired Research Lab, Ohio State University — *May 2023 - Present*

Graduate Teaching Associate — *Columbus, OH*
Department of Computer Science & Engineering, Ohio State University — *Aug 2023 - Present*

- Introduction to Neural Networks
- Software-*I*: Software Components (Course Instructor)
- Introduction to Computer Programming in Python

Design Engineer — *Islamabad, Pakistan*
Center for Advanced Research in Engineering — *Nov 2020 - May 2022*

Developer Intern — *Islamabad, Pakistan*
N-ovative Health Technologies — *July 2020 - Oct 2022*

Education

Ohio State University — *Columbus, OH*
Ph.D. in Computer Science & Engineering, GPA: 3.85 / 4.0 — *Aug 2023 - Present*

- Research: Neural Representations for Perceptual Speech Quality
- Advisor: Donald S. Williamson

National University of Sciences and Technology — *Islamabad, Pakistan*
BS in Electrical Engineering — *Sept 2016 - July 2020*

- Major: Signal Processing and Machine Learning
- Thesis: Real-time Adaptive Filtering for Robust Vibration Sensing

Projects

Lightweight AI for Perceptual Speech Quality on the Edge

- Lightweight attention-only model (86K params) for speech quality.
- 100× smaller than wav2vec2, yet comparable performance on SOMOS dataset.
- Self-teaching improves noisy-label robustness, $+8\%$ correlation.

One Model for All Domains: Zero-Shot Speech Quality Assessment

- Applied Sharpness-Aware Minimization (SAM) on unified MOS datasets.
- Promotes diverse features, reduces corpus effect, enhances cross-domain generalization.
- Up to 76% MSE reduction and 42% correlation improvement over Adam across 12 test sets.

Smartphone-Based Urine Test Strip Analysis for At-Home Diagnostics

- Smartphone-based system for accurate urine test strip color reading.
- Combines linear regression color calibration with Euclidean color matching.
- Achieved 92% accuracy in-house; low-cost, portable alternative to lab analyzers.

Sequential Pattern Mining for Hospital Fraud Detection

- Applied PrefixSpan to mine frequent patterns from 62 medical specialties.
- Patterns used in rule-based engine to assess transaction confidence and flag potential fraud.

Publications

A Generalization Strategy for Speech Quality Prediction: From Domain-Specific to Unified Datasets May 2026
Imran E Kibria, Ada Lamba, Donald S. Williamson
IEEE ICASSP 2026 (Under Review)

AttentiveMOS: A Lightweight Attention-Only Model for Speech Quality Prediction Aug 2025
Imran E Kibria, Donald S. Williamson
ISCA Interspeech 2025

Smartphone-Based Point-of-Care Urinalysis Assessment July 2022
Imran E Kibria, Hussnain Ali, Shoab A. Khan
IEEE EMBC 2022

Technologies

Languages: Python, MATLAB, Java, C, C++, SQL, LaTex.

Deep Learning: Tensorflow, PyTorch, HuggingFace, Github

Speech Toolkits: SHEET, ESPNet, SpeechBrain

Awards

CSE Scarlet & Gray Award Aug 2023
Merit-Based Scholarship, The Ohio State University (5-year award)

Best English Speaker June 2017
SEECS (School of Electrical Engineering & Computer Science) Declamation Contest

Certificate of Excellence Dec 2014
Third Place – Secondary Board Examination, Cadet College Hasanabdal